

Syllabus:

LA 470: Drone-aided Design

Landscape Architecture Department

Cal Poly-San Luis Obispo

Class Time: W 10am – 12pm

Class Room: 34-252 Lecture Room

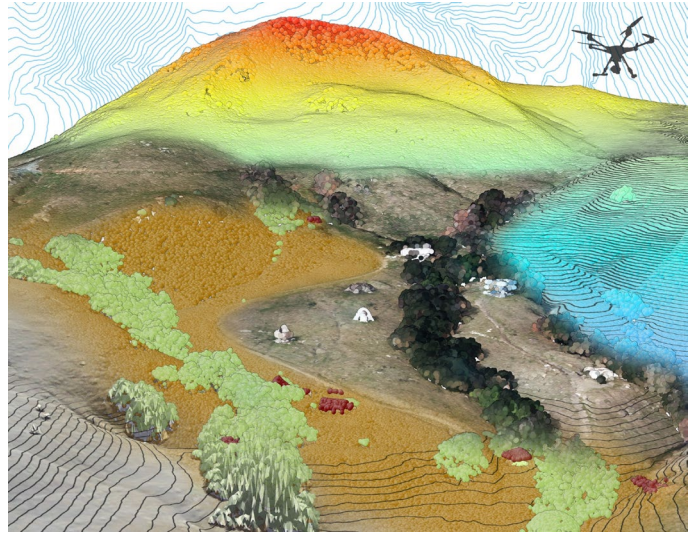
Instructor: Aaron Liggett

Office: Room 253 Dexter #34

Office Hours: M 12pm-1pm, F 12pm-1pm, or by appt.

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Cell: 805-895-2295



Course Description

Introduction to drone-based site data collection, analysis, and visualization. Covers flight operations, photogrammetry, GIS, 3D modeling, urban and ecological assessment, and simulation techniques to support concept development, planning, and performance-based design.

Course Description

Drone-aided design is a methodology that uses drone-based remote sensing to collect, analyze, and visualize spatial and environmental data, informing decision-making throughout the design process. It integrates aerial datasets—such as high-resolution imagery, 3D terrain models, thermal patterns, and vegetation indices—into workflows that connect site observation, analysis, modeling, and design development. Rather than serving solely as tools for documentation, drones function as analytical instruments that reveal otherwise unseen landscape conditions and dynamics.

This course introduces drones as analytical and design tools for the built and natural environment, integrating remote sensing, geospatial analysis, and 3D modeling into

design workflows. Through applied, project-based learning, students will engage workflows that connect data collection, analysis, and simulation to design decision-making across transdisciplinary practice and research. Students will develop both analog and digital skillsets, including ArcGIS Pro, Drone2Map, 3D modeling (Rhino, SketchUp), and physical model making, to follow established workflows and develop their own methods using drone technology. Applications span landscape architecture, architecture, construction management, and environmental and urban planning.

Course Learning Outcomes

Students will be expected to meet the following learning objectives:

- Understand drone systems, sensors, and flight planning
- Interpret datasets including orthomosaics, DEMs, point clouds, NDVI, and thermal imagery
- Analyze landscape conditions (topography, vegetation, hydrology, microclimate)
- Evaluate how drone data informs design decisions
- Integrate outputs into GIS and 3D modeling workflows

Text

No required text. Readings will be assigned and provided periodically throughout the quarter.

Course Structure

This course meets once per week for a two-hour session. Each class will begin with a lecture introducing key concepts, methods, and workflows. The remaining time will be dedicated to group discussions, hands-on activities, and applied exercises that reinforce lecture material and support project development.

The course will include guest lectures from professionals and researchers providing additional perspectives on emerging tools and practices, which, on occasion, will be held virtually.

Students are expected to arrive on time and attend all class sessions. Important announcements, lectures, and course communications will be delivered at the start of class, and active participation is essential to the learning experience

Assignments

The project phases (assignments) and grade weightings are:

Project Phase	% Weight of Total Grade
Weekly Assignments	50%
Final Project	35%
Participation	15%

Participation

Participation is critical for success in this class since it is a hands-on course. Participation is evaluated by class attendance, arriving to class on time, staying for the duration of the class period, being prepared for class discussions, and completion of in-class exercises and homework. Absence from class and the failure to complete assignments will result in deductions from the participation grade. 15% of the course grade is based on students' attendance, punctuality, and participation in class, including staying until the end of class.

Late/Missed Work and Make-Up Policy

Late and missed work may be allowed in the case of excused absences (see attendance policy below) and on a case-by-case basis. Please see me if you anticipate difficulty in submitting your work in a timely manner. Late and missed work must be completed and turned in as soon as possible. Late work is turned in that is not the result of an excused absence; it will be penalized 20% for the first day and 5% for each day thereafter.

Grading Policy

Plus/Minus grading will be in effect for the quarter. The following breakdown of scores and percentages will determine your final grade in the course:

A	100% to 93%
A-	< 93% to 90%
B+	< 90% to 87%
B	< 87% to 83%
B-	< 83% to 80%
C+	< 80% to 77%
C	< 77% to 73%
C-	< 73% to 70%
D+	< 70% to 67%
D	< 67% to 63%
D-	< 63% to 60%
F	< 60% to 0%

Grades will be based on the following criteria:

A – Work reflects achievement that is outstanding relative to the level necessary to meet the course objectives

B – Work reflects achievement that is significantly above the level necessary to meet the course objectives

C – Work reflects acceptable attainment of the course objectives

D – Work reflects poor attainment of the course objectives

F – Work reflects non-attainment of the course objectives

Course Expenses

Books and required text:	free (uploaded in Canvas or online at Kennedy library)
Printing & model material:	\$10-\$30

Course Schedule

(Calendar is subject to change)

1	04/01 Course Intro Lecture: Introduction to Drone Aided Design
2	04/08 Guest Lecture: Dr. Marc Horney, Professor (Rangeland Ecology and Management)
3	04/15 Lecture: Flight Planning and Photogrammetry
4	04/22 Field Drone Flight
5	04/29 VIRTUAL CLASS Guest Lecture: Eric Arneson (Landscape Architect)
6	05/06 VIRTUAL CLASS Guest Lecture: ALTHOUSE and MEADE (Environmental Services)
7	05/13 Lecture: GIS Analysis and Digit Modeling
8	05/20 Lecture: Performance Analysis and Design
9	05/27 Lecture: Visualizations and Simulations
10	06/03 Final Presentation

University Policies

Attendance

- You are expected to be present and working during class period (8:10am – 12:00 pm) unless class is dismissed early
- Working on other projects or course material during class time is not allowed.
- Absences are acceptable for the following reasons: illness, unforeseen incidents such as an accident, serious illness or death of a close relative, religious holidays, NCAA competitions (with letter from the coach or athletic department), active participation in university events (with a letter from the advisor), field trips for other classes (with a letter from the professor), selective service/military reasons. ***You are not excused from the work during absences*** and it is your responsibility to determine what information and work was missed and to make it up.

Inclusion and Classroom Climate

I strive to make this classroom a place where you will be treated with respect, and I welcome individuals of all ages, backgrounds, beliefs, races, ethnicities, social classes, genders, gender identities, gender expressions, national origins, documentation statuses, religious affiliations, sexual orientations, abilities – and other visible and nonvisible differences. All members of this class are expected to contribute to a respectful and inclusive environment for every other member of the class. This does not mean we cannot disagree or have different ideas. It does mean we try to consider perspectives other than our own, though they may differ from our own beliefs/experiences. If you experience disrespect or discrimination in this class, please report your experiences to me.

Academic Integrity

Cal Poly is dedicated to nurturing an inclusive and academically excellent community, which can only flourish with honesty, trust, and mutual respect. At the core of academic excellence is the search for knowledge through the open and dynamic exchange of ideas. You and your student peers must have a strong commitment to personal and professional integrity that informs your behavior both before and after graduation, discouraging you from creating a false appearance of achievement by presenting the work of others as your own, or bending or breaking the rules of any situation. Either action can lead to serious consequences, academic and otherwise. Please thoroughly familiarize yourself with the Office of Students' Rights and Responsibilities page on Academic Integrity, as well as the information found on Academic Programs and Planning website regarding cheating and plagiarism.

Academic Support for Students

The Tutoring & Learning Center (TLC) provides free academic support to all Cal Poly students. Supporting more than 250 courses and subjects across all six colleges, the TLC offers a collaborative space, both in person and online, for students to share learning strategies and work toward meeting the expectations of course assignments, projects, exams, and essays. On average, the TLC hosts nearly 15,000 student consultations each academic year. Seeking help through learning support services like tutoring, study sessions, and supplemental workshops available through the TLC is a positive behavior that will assist you in attaining academic success. For more information, visit the TLC website or stop by the Center in Kennedy Library, Room 209.

Alcohol and Drug Abuse Prevention

Many students are impacted by alcohol and drug use while in college, and I want to make sure you have the support you need to thrive personally and academically. Wellbeing Services offers a variety of alcohol and drug resources to support students in making safer, healthier decisions. If you have any concerns about your substance use, I encourage you to reach out and utilize the services below:

<https://chw.calpoly.edu/counseling/alcohol-and-other-drug-resources>

Basic Needs Support

If you don't have stable housing and/or are skipping meals, are concerned about spending money on food, and/or have difficulties accessing healthy food options, learn about the many resources available to support you through Cal Poly's Basic Needs Initiative at basicneeds.calpoly.edu. We can provide you with immediate food, housing and emergency funding support and access to additional resources.

If I can help you in any way to access the resources above, or if you have any questions about student care resources, please contact me so that I can assist you. I am committed to ensuring that all students have the resources they need to be able to participate in this course.

Disability Support Resources

It is University policy to provide, on a flexible and individualized basis, reasonable accommodations to students who have disabilities that may affect their ability to

participate in course activities or to meet course requirements. If you have a disability for which you are or may be requesting an accommodation, you are encouraged to contact both your instructor and the Disability Resource Center, Building 124, Room 119, at (805) 756-1395, as early as possible in the term. Apply for Disability Services or email drc@calpoly.edu.

Pronouns and Preferred Names

At Cal Poly everyone has the right to be addressed by their preferred name and the personal pronouns that correspond to their gender identity. Students can update their pronouns in PolyProfile and in their Canvas user settings. If you have not yet updated your pronouns in Canvas, you can do so at the beginning of the term so that I can make sure to refer to you using the correct pronouns. If your pronoun set is not available in Canvas, please let me know in the survey that I will distribute to students. I will survey all students in this course so that you can indicate the name and pronouns you would like me to use when addressing you in this course.

As part of our commitment to equity and inclusion in this course, it is important that all students in this class respect the pronouns and preferred names of their peers. Mistakes in addressing one another may happen. If you make a mistake or are corrected, please briefly apologize and correct yourself. I will not tolerate intentional disrespect or discrimination towards other students based on their pronouns, preferred name, gender identity, or any other social identity. I recognize that preferred names and personal pronouns may change during the course of the quarter, if at any point during the class you would like to be addressed differently, please let me know. To learn more about personal pronouns and why they are important please visit Cal Poly's Pronouns Matter website and mypronouns.org.

Students with Dependents

One in five college students are parents. Student Parents earn higher GPAs than their non-parenting peers. Cal Poly, San Luis Obispo has a Students with Dependents program. These facts are presented to support the experience of parenting, pregnant, and caregiving students on campus.

If you are a Student with Dependents, or know of anyone who is, please connect with the Students with Dependents program via email at studentswithdependents@calpoly.edu and or on the Students with Dependents website.

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Wellbeing Support

Recent nationwide surveys of college students consistently find that stress, sleep problems, anxiety, depression, interpersonal concerns, death of a significant other and alcohol use are among the top ten health impediments to academic performance. Students experiencing personal problems or situational crises are encouraged to contact Cal Poly's Counseling Services (805-756-2511) for assistance, support and advocacy. This service is free and confidential.